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An improved Otsu method for oil spill detection from SAR images

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Summary In recent years, oil spill accidents have become increasingly frequent due to the development of marine transportation and massive oil exploitation. At present, satellite remote sensing is the principal method used to monitor oil spills. Extracting the locations and extent of oil spill spots accurately in remote sensing images reaps significant benefits in terms of risk assessment and clean-up work. Nowadays the method of edge detection combined with threshold segmentation (EDCTS) to extract oil information is becoming increasingly popular. However, the current method has some limitations in terms of accurately extracting oil spills in synthetic aperture radar (SAR) images, where heterogeneous background noise exists. In this study, we propose an adaptive mechanism based on Otsu method, which applies region growing combined with both edge detection and threshold segmentation (RGEDOM) to extract oil spills. Remote sensing images from the Bohai Sea on June 11, 2011 and the Gulf of Dalian on July 17, 2010 are utilized to validate the accuracy of our algorithm and the reliability of extraction results. In addition, results according to EDCTS are used as a comparator to further explore validity. The comparison with results according to EDCTS using the same dataset demonstrates that the proposed self-adapting

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algorithm is more robust and boasts high-accuracy. The accuracy computing by the adaptive algorithm is significantly improved compared with EDCTS and threshold method.

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1. Introduction

Oil spill events have caused serious damage to the ocean environment in the past decade. Although the magnitude of most oil spills is less than 7 tons, large spills over 700 tons are likely to degrade and destroy a variety of marine biodiversity and perturb ecosystem functioning according to data from International Tanker Owners Pollution Federation Limited (Ng and Song, 2010). Oil spill accidents have captured people's attention and should be taken more seriously.

At present, there are two methods for oil spill monitoring, SAR and optical method. Remote sensing is considered as an effective method to detect oil spills. Among satellite sensors, the optical method is useful to detect oil floating below the sea surface in terms of the changes of wavelength and sea surface reflectance (Otremba, 2016). Meanwhile, SAR is the most useful for detecting oil spills due to its wide area coverage and day/night capabilities (Garcia-Pineda et al., 2009; Liu et al., 2011; Migliaccio et al., 2012; Zhang et al., 2011). With the increasingly extensive and in-depth application of quantitative SAR images, there is a greater onus than ever placed on the remote sensing community to optimize extraction procedures. Indeed, bespoke algorithms have been developed and presented by scientists from a wide variety of different institutions. As in the earlier study, scientists researched one hundred images of offshore areas in Europe and the seasonality of oil spill patterns was summarized (Gade and Alpers, 1999; Solberg et al., 1999). The results of extraction were crude and this was a time-consuming endeavor. The early classical and effective method (Otsu) is utilized which was proposed by Otsu in 1979 (Ng, 2006; Otsu, 1979; Xu et al., 2011). For the purposes of increasing accuracy, some researchers proposed an improved and merging algorithm for the extraction of oil, which although boasting performance benefits in terms of accuracy, the threshold is single and cannot adapt to variable pixels (Mehnert and Jackway, 1997; Tremeau and Borel, 1997). In the early 20th century, scientists developed an automatic classifier for oil spill detection. A model to automate the identification of oil was operationalized and it could capture range and directional parameters by simulating the diffusion of oil (Marghany, 2001, 2004). Due to the variety of different water areas, a fast region-based detection algorithm for oil spill detection in SAR images was proposed which used a new image segmentation technique. An oil spill model was developed and employed generalized likelihood ratio test (GLRT) detection theory to improve the accuracy of oil spill detection; but when a large number of SAR images must be examined, this could be a labor-intensive task (Chang et al., 2008; Huo et al., 2010). In order to militate against these problems, a modified constant false alarm rate (CFAR) was proposed which combined an edge

detection technique and CFAR theory to improve accuracy (Vikhe and Thool, 2016).

However, simple edge detection and the adaptive threshold method cannot extract spills information accurately enough; the boundary of certain oil slicks may not be closed by using simple edge detection; which caused the loss of oil information. Meanwhile, threshold segmentation may mistake some regions influenced by wind and waves, as oil spills. In this paper, we add region growing which can better recognize the gray similarity between one point and the nearby points. This method can offer greater reliability in terms of representing oil film edge information. In order to quantify the reliability of the algorithm, data pertaining to another oil spill event are utilized. Further, the oil films extracted from remote sensing images of the Penglai oil field on June 11, 2011, the Gulf of Dalian on July 17, 2010 and the North Seas on April 11, 1994 by the new algorithm are compared with results according to EDCTS and threshold method.

This paper is structured as follows. The remote sensing images and the method of extracting oil spill information are described (Section 2) followed by the results of the analysis with a focus on discussing the advantages of the new algorithm proposed herein (Section 3). Finally, some conclusions are offered which include suggestions for future research (Section 4).

2. Material and methods

2.1. Dataset

According to our requirements and recognizing the necessity of validating this new algorithm, two sets of remote sensing images are deployed in the study. The remote sensing images are acquired from the ENVISAT-1 satellite, which was launched by the European Space Agency (ESA) on March 1, 2002, and carries an active microwave instrument advanced synthetic aperture radar (ASAR). The ASAR instrument is a multi-mode sensor, which operates at C-band (i.e., center frequency of 5.3 GHz) at several polarizations (HH, VV, HV and VH), incidence angles, and in the following functioning modes: image mode, polarization mode, wide swath mode, global monitoring mode, and wave mode. The remote sensing image of the Bohai Sea (Fig. 1), the Gulf of Dalian (Fig. 2) and the North seas (Fig. 3) are both presented.

2.2. Algorithm for oil spill identification

Preprocessing and image segmentation are the most important stages for oil recognition. The grayscale image is divided

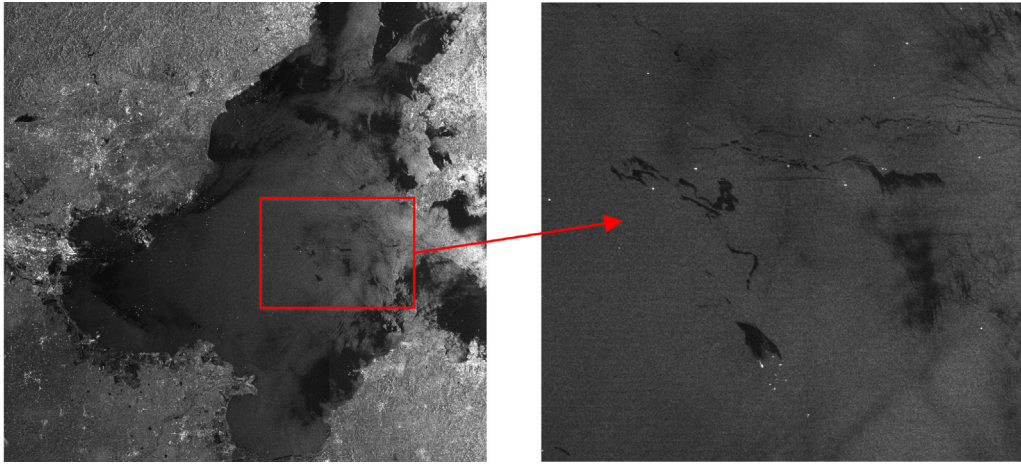


Figure 1 Slick positions observed by ENVISAT-1 ASAR on June 11, 2011, of the Bohai Sea.

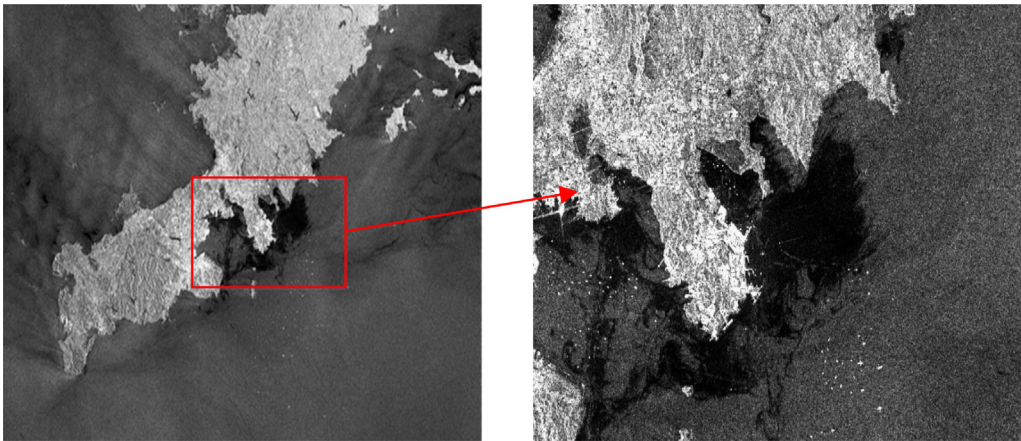


Figure 2 Slick positions observed by ENVISAT-1 ASAR on July 17, 2010, of the Gulf of Dalian.

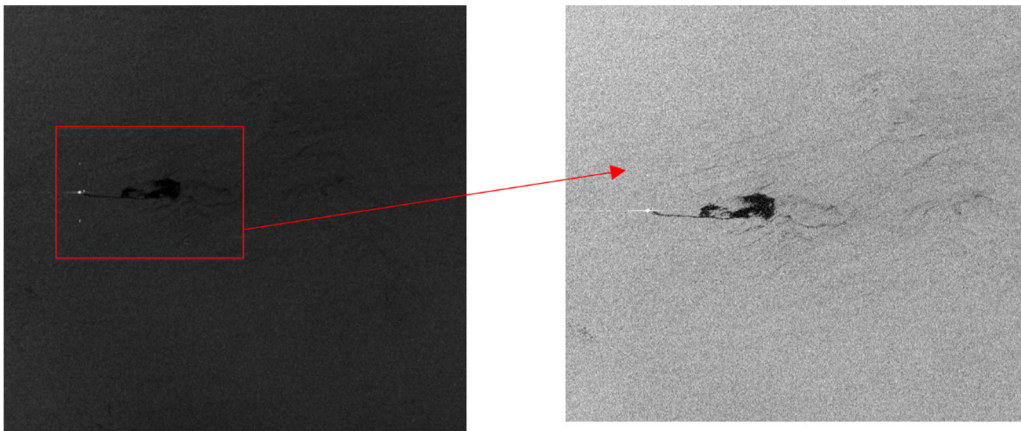


Figure 3 Slick positions observed by remote satellite on April 11, 1994, of the North Seas.

into two parts, target and background. A gray image is described as $F_{M \times N} = [f(x, y)]_{M \times N}$ using a two-dimensional matrix. $M \times N$ is the size of the image, and $f(x, y) \in \{0, 1, \dots, L-1\}$ is the gray value of the pixel (x, y) . L is the total number of gray levels. The number of gray level i appearing in

the image is n_i ; the probability of the emergence of gray level i is as follows:

$$p_i = \frac{n_i}{M \times N}, \quad p_i \geq 0, \quad \sum_{i=0}^{L-1} p_i = 1. \quad (1)$$

Suppose that the gray level t is the threshold of the image gray value, and the total pixels are divided into two parts: BG (background) contains $i \leq t$ pixels, FG (foreground) contains $i > t$ pixels. The odds of BG and FG are as follows:

$$\begin{cases} P_1 = \sum_{i=0}^t p_i, \\ P_2 = \sum_{i=t+1}^{L-1} p_i, \\ P_1 + P_2 = 1. \end{cases} \quad (2)$$

In order to describe the distinction between two classes effectively, we define D as the class difference of BG and FG, respectively:

$$\begin{cases} \omega_1 = \sum_{i=0}^t i p_i / P_1, \\ \omega_2 = \sum_{i=t+1}^{L-1} i p_i / P_2, \\ D = |\omega_1 - \omega_2|. \end{cases} \quad (3)$$

D is a defined extent which can reflect the BG and FG segmentation effect. The greater the D the greater the distance between the two classes, BG and FG are more divergent. In addition, the cohesion of the two classes is also a direct indicator of the effectiveness of the partition; the distance between each pixel in the class is defined as the degree of dispersion of the class:

$$\begin{cases} d_1 = \sum_{i=0}^t |i - \omega_1| \cdot \frac{p_i}{P_1}, \\ d_2 = \sum_{i=t+1}^{L-1} |i - \omega_2| \cdot \frac{p_i}{P_2}. \end{cases} \quad (4)$$

Obviously, the smaller the degree of dispersion of each class, the better the resulting classification. Therefore, to ensure adequate classification we must ensure that D is maximum and d_1, d_2 are minimums. In this way, class cohesion is optimal, and the class difference is the largest. The definition of the BG and FG classification discriminant function is as follows:

$$H(t) = \frac{P_1 \cdot P_1 \cdot D}{P_1 \cdot d_1 + P_2 \cdot d_2}. \quad (5)$$

The best classification results will be achieved when H is maximum. We set FG pixel gray value to 0, and BG pixel gray value to 1. Next, we combine threshold segmentation with edge detection. At present, the canny operator is considered as the most effective method in edge detection (Cai et al., 2015). However, the boundary of some oil slicks may not be closed by using simple edge detection, which will result in the missing of oil slicks when multiplied by the result of threshold segmentation. In order to increase the accuracy of extraction, we add region growing to this for the purposes of edge detection when integrated with the result of image segmentation.

The approach of region growing for seeded regions is to segment an image into regions with a set of n seed regions

(Adams and Bischof, 1994). Each seed region is a connected component of one or more pixels and represented by a set of A_i , where $i = 1, 2, \dots, n$. T stands for the set of all unallocated pixels and the border is one of the A_i , i.e.,

$$T = x \in \bigcup_{i=1}^n A_i : N(x) \cap \bigcap_{i=1}^n A_i \neq \emptyset, \quad (6)$$

where $N(x)$ represents the set of immediate neighbors-8 for the square grid of the pixel of threshold segmentation's result. A single step of the algorithm involves examining the neighbors of each x belonging to T in turn; if $N(x)$ intersects a region A_j , then calculating the difference (similarity) between x and the intersected region. In the simplest case $\delta(x)$ is defined as follows:

$$\delta(x) = |g(x) - \text{mean}_{y \in A_j} \{g(y)\}|, \quad (7)$$

where $g(x)$ is the intensity (gray value) of the pixel x . If $N(x)$ intersects more than one region, then A_j is taken to be that region for which $\delta(x)$ is a minimum (alternatively, the pixel x can be flagged as a boundary pixel for display purposes). In this way, δ value is determined for each $x \in T$. Finally, the pixel $z \in T$ is calculated as the function (13). This process continues until all of the image pixels have been assimilated.

$$\delta(z) = \min_{x \in T} \{\delta(x)\}. \quad (8)$$

3. Results and discussion

3.1. Information extraction

The purpose of this study is focused on extracting oil information in SAR images. Fig. 3 shows the rough edge of oil detected by edge detection and the rough region of oil detected by threshold segmentation. It is obvious that the oil edge of region 1 in Fig. 4(a) is not closed, which will become problematic if remedial action is not taken. The uncertainty associated with the threshold segmentation results is significant, especially in the context of region 3 which illustrates non-detection issues with the edge detection method (Fig. 4b). Fig. 4(a) shows the result of combining simple edge detection and single threshold segmentation. It is clear that two regions of oil are lost and many small hot

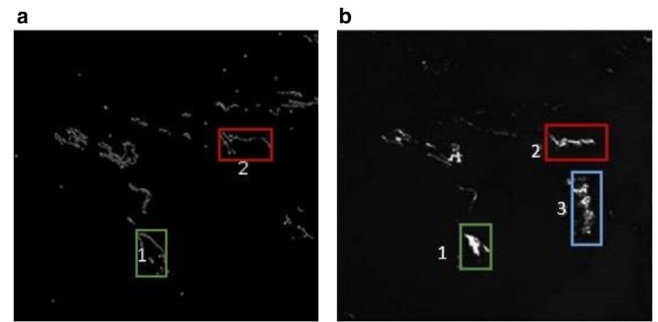


Figure 4 The rough edge of oil detected by (a) edge detection, (b) threshold segmentation.

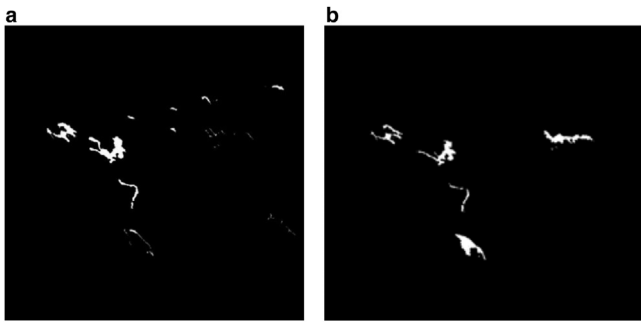


Figure 5 The result of using (a) EDCTS and (b) RGEDOM as applied to the Bohai Sea.

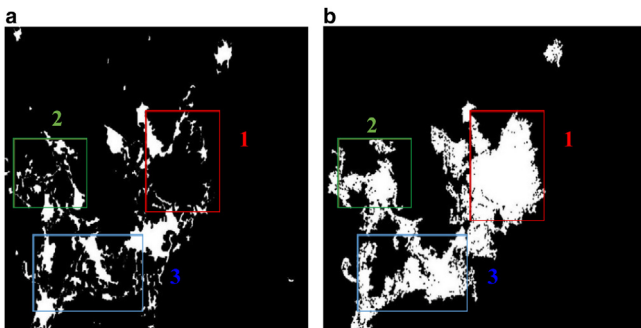


Figure 6 The result of using (a) EDCTS and (b) RGEDOM as applied to Dalian.

spots are misclassified, possibly due to the wind, wave, and other influencing factors.

In this study, the sub-regions are combined through traversing eight directions based on the regions selected by threshold segmentation. The optimal closed path will be found by repeated convolution. Then, the latest oil slick boundary is used to match the region extracted by threshold segmentation to arrive at the final result. The extracted oil information according to this new method is shown in Fig. 5 (b). Comparing this with Fig. 4 we can identify that the method retained open region 1 whilst removing the redundant region induced by the wind, wave, and sea surface topography. In sum, there are five oil spots which accord well with officially documented oil slick data.

To further explore the robustness of the new algorithm, an additional image is utilized. Compared with the previous image, image texture is more complex and edge definition is blurrier. Further, the land boundary increases the difficulty of oil film recognition and is thus more demanding of the

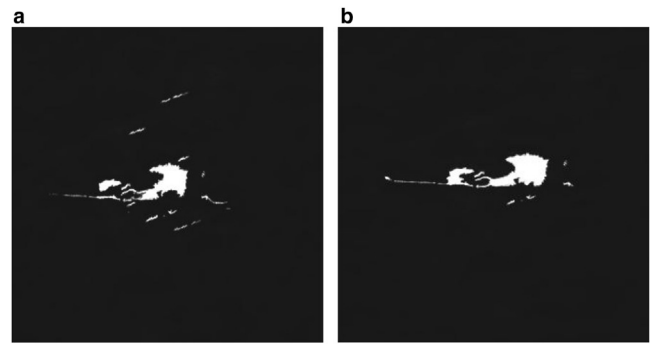


Figure 7 The result of using (a) EDCTS and (b) RGEDOM as applied to North Seas.

algorithm. The oil films extracted by EDCTS and RGEDOM are shown in Fig. 6(a) and (b) respectively. From Fig. 6(a) we can see that the majority of the edge in regions 1 and 3 is not closed well due to the blurred oil film boundary in the remote sensing image. Region 2 in Fig. 6(a) is adjacent to the land boundary are adjacent, leading to unclosed edge issues. The new algorithm addresses non-closed boundary issues, which greatly improves the accuracy of the extraction. The edge of the three regions in Fig. 6(b) is well closed and consistent with the original image.

Another SAR image of an oil spill in the North Sea is shown in Fig. 7. The oil films extracted by EDCTS and RGEDOM are shown in Fig. 7(a) and (b) respectively. From Fig. 7, we can see that the accuracy has a significant improve using the RGEDOM method compared to EDCTS by means of interpretation.

3.2. Quantitative analysis

In order to prove the effectiveness of the method, we conduct a quantitative analysis of the results that getting by the different algorithms. The results are shown in Fig. 8, the area of the oil spill and error rate (ER) are shown in Table 1. From Fig. 8, we can see that the new algorithm can solve the non-closed boundary problem well. The result getting by the RGEDOM method is more consistent with the original image than the two additional methods. Further, Table 1 demonstrates that the error rate has reduced significantly and the accuracy of oil spill area has improved obviously. In summary, the new algorithm is time-saving and high accuracy in ordinary remote-sensing image processing. The extraction results can play an important role in forecasting of oil spill accidents.

Table 1 The oil spill area and error rate.

Location	Real	EDCTS	Threshold method	Otsu
Penglai oil field [km ²] error rate	39.85	45.42	51.76	40.93
	—	14.0%	30.0%	3.0%
Gulf of Dalian [km ²] error rate	430	322.51	645.12	486.65
	—	25.0%	50.0%	13.0%

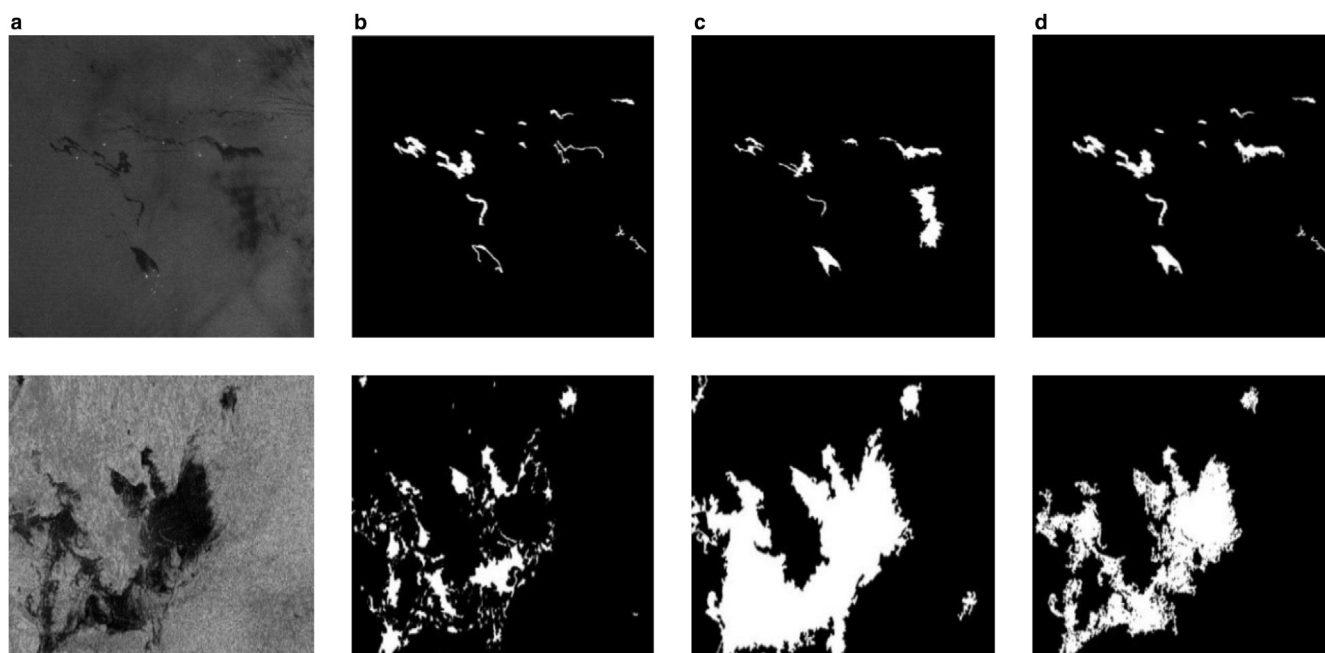


Figure 8 Comparison of results across three different methods (a) original image, (b) EDCTS, (c) threshold method, (d) RGEDOM.

4. Conclusions

The exact location of oil spills is essential and meaningful for simulating of the trajectory. The RGEDOM method presented in this paper is based on region growing and adaptive threshold segmentation. Experiments demonstrated that the RGEDOM method can work more effectively than comparator detectors for oil spill contexts. The SAR images are heterogeneous, although oil slicks are usually characterized by dark coloration, such coloration is not always synonymous with oil spill areas, instead, it could reflect the influence of the wind, wave and other factors. This method aimed to detect and approximate oil based on threshold segmentation and edge detection. Then, using region growing to match the results of threshold segmentation from eight directions centers on the location of the oil texting by edge detection, which can produce results which are more accurate and reliable.

Nevertheless, there are further influencing factors which require investigation, particularly disruptive factors should be considered. Finally, in future, this newly developed method should be auto-adjusted to adapt to the different climate background of oil spills and the multiple manifestations of ocean noise.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at [doi:10.1016/j.oceano.2017.03.005](https://doi.org/10.1016/j.oceano.2017.03.005).

References

- Adams, R., Bischof, L., 1994. Seeded region growing. *IEEE Trans. Pattern Anal.* 16 (6), 641–647, <http://dx.doi.org/10.1109/34.295913>.
- Cai, L., Ma, Y., Yuan, T., Wang, H., Xu, T., 2015. An application of canny edge detection algorithm to rail thermal image fault detection. *J. Comput. Comm.* 3 (11), 19–24, <http://dx.doi.org/10.4236/jcc.2015.311004>.
- Chang, L., Tang, Z., Chang, S.H., Chang, Y.L., 2008. A region-based GLRT detection of oil spills in SAR images. *Pattern Recogn. Lett.* 29 (4), 1915–1923, <http://dx.doi.org/10.1016/j.patrec.2008.05.022>.
- Gade, M., Alpers, W., 1999. Using ERS-2 SAR images for routine observation of marine pollution in European coastal waters. *Geosci. Remote Sens. Symp. Proc.* 237 (1999), 441–448, [http://dx.doi.org/10.1016/S0048-9697\(99\)00156-4](http://dx.doi.org/10.1016/S0048-9697(99)00156-4).
- Garcia-Pineda, O., Zimmer, B., Howard, M., Pichel, W., Li, X., MacDonald, I.R., 2009. Using SAR images to delineate ocean oil slicks with a texture-classifying neural network algorithm (TCNNA). *Can. J. Remote Sens.* 35 (5), 411–421, <http://dx.doi.org/10.5589/m09-035>.
- Huo, Y.K., Wei, G., Zhang, Y.D., Wu, L.N., 2010. An adaptive threshold for the Canny Operator of edge detection. In: 2010 International Conference on Image Analysis and Signal Processing, Zhejiang, 371–374, <http://dx.doi.org/10.1109/IASP.2010.5476095>.
- Liu, P., Li, X., Qu, J.J., Wang, W., Zhao, C., Pichel, W., 2011. Oil spill detection with fully polarimetric UAVSAR data. *Mar. Pollut. Bull.* 62 (12), 2611–2618, <http://dx.doi.org/10.1016/j.marpolbul.2011.09.036>.
- Marghany, M., 2001. RADARSAT automatic algorithms for detecting coastal oil spill pollution. *Int. J. Appl. Earth Obs.* 3 (2), 191–196, [http://dx.doi.org/10.1016/S0303-2434\(01\)85011-X](http://dx.doi.org/10.1016/S0303-2434(01)85011-X).

- Marghany, M., 2004. RADARSAT for oil spill trajectory model. *Environ. Modell. Softw.* 19 (5), 473–483, [http://dx.doi.org/10.1016/S1364-8152\(03\)00162-2](http://dx.doi.org/10.1016/S1364-8152(03)00162-2).
- Mehnert, A., Jackway, P., 1997. An improved seeded region growing algorithm. *Pattern Recogn. Lett.* 18 (10), 1065–1071, [http://dx.doi.org/10.1016/S0167-8655\(97\)00131-1](http://dx.doi.org/10.1016/S0167-8655(97)00131-1).
- Migliaccio, M., Nunziata, F., Brown, C., Holt, B., Li, X., Pichel, W., Shimada, M., 2012. Polarimetric synthetic aperture radar for transocean Deepwater Horizon oil accident monitoring. *Eos* 16 (93), 161–168.
- Ng, H.F., 2006. Automatic thresholding for defect detection. *Pattern Recogn. Lett.* 27 (14), 1644–1649, <http://dx.doi.org/10.1016/j.patrec.2006.03.009>.
- Ng, A.K.Y., Song, S., 2010. The environmental impacts of pollutants generated by routine shipping operations on ports. *Ocean Coast. Manage.* 53 (5–6), 301–311, <http://dx.doi.org/10.1016/j.ocecoaman.2010.03.002>.
- Otremba, Z., 2016. Oil droplet clouds suspended in the sea: can they be remotely detected. *Remote Sens.* 8 (10), 857–865, <http://dx.doi.org/10.3390/rs8100857>.
- Otsu, N., 1979. A threshold selection method from gray-level histograms. *Automatica* 11 (285–296), 23–27, <http://dx.doi.org/10.1109/TSMC.1979.4310076>.
- Solberg, A.H., Storvik, G., Solberg, R., Volden, E., 1999. Automatic detection of oil spills in ERS SAR images. *IEEE Trans. Geosci. Remote Sens.* 37 (4), 1916–1924, <http://dx.doi.org/10.1109/36.774704>.
- Tremeau, A., Borel, N.A., 1997. A region growing and merging algorithm to color segmentation. *Pattern Recogn. Lett.* 30 (7), 1191–1203, [http://dx.doi.org/10.1016/S0031-3203\(96\)00147-1](http://dx.doi.org/10.1016/S0031-3203(96)00147-1).
- Vikhe, P.S., Thool, V.R., 2016. Mass detection in mammographic images using wavelet processing and adaptive threshold technique. *J. Med. Syst.* 40 (4), 1–16, <http://dx.doi.org/10.1007/s10916-016-0435-3>.
- Xu, X., Xu, S., Jin, L., Song, E., 2011. Characteristic analysis of Otsu threshold and its applications. *Pattern Recogn. Lett.* 32 (7), 956–961, <http://dx.doi.org/10.1016/j.patrec.2011.01.021>.
- Zhang, B., Perrie, W., Li, X., Pichel, W.G., 2011. Mapping sea surface oil slicks using RADARSAT-2 quad-polarization SAR image. *Geophys. Res. Lett.* 38 (10), 602–606, <http://dx.doi.org/10.1029/2011GL047013>.